

Matheus Rosa

AI/ML Engineer · GenAI & MLOps · Recife, Brazil · Remote (US/EU overlap)

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English: C1 (Advanced)

SUMMARY

ML Engineer and Data Engineer, 4+ years with Python and AWS, 2 years with generative and multimodal AI. I build the production systems around them: Terraform infrastructure-as-code, CI/CD for models, Airflow orchestration, Databricks pipelines, MLflow observability. Strong in data pipeline development and data modeling for Data Lake and Data Lakehouse environments (Databricks, PySpark, Delta Lake). Right now I own LLM evaluation and observability on Databricks and MLflow: judge harnesses, tool-calling accuracy suites, per-request cost attribution. Computer vision research background at UFPE. Most of that work has been in Agile teams, often as the only engineer on the problem, so the data pipelines and the backend were mine too.

EXPERIENCE

- **Maggu** **Remote**
Data AI Engineer (company wide layoff by cost reductions) *Apr. 2026 – Aug. 2026*
 - Medicine recommendation accuracy went from 62% to 94% by reworking LLM judge from a naive judge to two adversarial agents backed up by human made dataset on mlflow hosted at databricks.
 - Tracked all LLM operational costs, latency, accuracy and its output routing to databricks mlflow, reworking cache and rag strategies cutting from 1.5 seconds to 0.7 seconds time to first token.
 - Provisioned the Databricks pipelines as infrastructure-as-code with Terraform, so every environment is reproducible from version control.
 - Reduced postgres daily overloads and reducing time data upload from 2 hours to 15 minutes by batching data and setting to 8 uploading threads.
 - Cut Databricks compute spend from 3.21\$ hourly dbu to 0.80\$ dbu by profiling PySpark jobs and fixing bad partitioning, joins that should have been broadcast and small-file churn.**Stack:** Databricks, PySpark, Delta Lake, MLflow, Terraform, Agno, Github actions
- **Capgemini** **Recife, Brazil (Hybrid)**
Data AI Engineer (3-month temporary contract) *Jan. 2026 – Apr. 2026*
 - Wrote the evaluation suite for agent workflows: does it call the right tool, with the right arguments, in the right order. Caught 4 regressions before they reached the client.
 - Ran workshops on agent design and LLM-as-a-judge for a newly formed team and 2 managers from Vivo, accelerating adoption of shared evaluation metrics.
 - Handled client technical communication in an Agile delivery team and mentored two data managers on agent design and evaluation.**Stack:** Python, LLM evaluation, agent frameworks, tool-calling validation, RAG, Agile/Scrum
- **Triggo.ai** **Remote**
Data AI Engineer (3-month temporary contract; invited to join Capgemini) *Dec. 2025 – Jan. 2026*
 - Engineered prompts to keep LLM tone consistent using self-consistency and tree-of-thoughts techniques.
 - Tuned Anthropic and OpenAI API calls to enforce JSON-formatted, length-constrained outputs.**Stack:** Python, Bruno, DocumentDB
- **Oncase** **Recife, Brazil (Hybrid)**
Data Engineer (3-month temporary contract) *Aug. 2025 – Nov. 2025*
 - Maintained a Flask API for image processing, storing uploads on S3 for a computer vision inference pipeline.
 - Maintained Airflow DAGs using a Gitflow branch pattern, keeping broken DAGs from ever hitting production.
 - Hosted Airflow on EC2 to run continuous inference and training jobs for the computer vision pipeline.**Stack:** Python, Flask, PaddleOCR, Airflow, AWS, Terraform
- **Acaso** **Recife, Brazil**
Machine Learning Engineer *Jan. 2025 – Jun. 2025*
 - Automated document ingestion in Airflow and gave the team back roughly 20% of its week.
 - Built the RAG pipeline: Docling for PDF parsing, LangChain for orchestration, PostgreSQL with pgvector for storage.
 - Cut manual data prep 60% by moving S3 ingestion into Python and Boto3 instead of hand-pulled exports.
 - Shipped generative AI proofs-of-concept in Streamlit and FastAPI so clients could try an idea before anyone committed to building it.**Stack:** Python, LangChain, PostgreSQL/pgvector, Docling, Apache Airflow, AWS S3, Boto3, FastAPI, Streamlit

- **Di2win**

Recife, Brazil

Machine Learning Engineer

Apr. 2024 – Jan. 2025

- Cut energy consumption 20% with LSTM and regularized regression forecasts tuned in Optuna. SARIMA was the baseline; it lost.
- Replaced manual retraining with event-driven Airflow pipelines that fire on data refresh and on drift.
- Set up GitHub Actions with Pytest coverage on model behavior and input data, not only on code.
- Built a containerized stack where model training, fine-tuning experiments and retraining ran isolated from live projects.

Stack: Python, PyTorch, Optuna, Apache Airflow, Docker, GitHub Actions, Pytest, FastAPI

- **Valorian**

Recife, Brazil (Remote)

Data Scientist

Apr. 2023 – Apr. 2024

- Reduced hosting costs 25% by containerizing workloads and moving to reserved instances.
- Automated ingestion and transformation end to end. That was 30 hours a month of manual work gone.
- Improved wheat flour production efficiency 12% with XGBoost models tuned in Optuna.

Stack: Python, XGBoost, Optuna, Docker, AWS EC2/S3, Dash/Plotly

TECHNICAL SKILLS

- **LLM & Agents:** LLM-as-a-judge evaluation, tool-calling and trajectory evals, RAG, LangChain, MCP, ChromaDB, pgvector, trace observability, token and cost attribution
- **MLOps & Infrastructure:** Docker, Terraform (IaC), Kubernetes, GitHub Actions, MLflow, Apache Airflow, Prefect, CI/CD for models, model registry, drift monitoring
- **Cloud:** AWS (ECS, EC2, S3, Lambda, ECR, CloudWatch, IAM, Boto3, SageMaker), GCP (Vertex AI, Cloud Run, BigQuery, GCS), Azure OpenAI
- **Data Engineering:** Databricks (PySpark, Delta Lake, Unity Catalog), PostgreSQL, Apache Kafka, Pandas, medallion architecture, idempotent MERGE pipelines
- **ML:** PyTorch, computer vision, scikit-learn, XGBoost, Optuna, LSTM, SARIMA, SciPy
- **Languages, APIs & Practices:** Python, SQL, Bash, FastAPI, Pydantic, Pytest, Agile/Scrum

EDUCATION

- **Center of Informatics, Federal University of Pernambuco (UFPE)**

Recife, Brazil

B.Sc. in Computer Science

July. 2022 – July 2026